

URALSK, Kazakhstan

WMO: 351080

Lat: 51.151N

Lon: 51.543E

Elev: 125

StdP: 14.63

Time Zone: 5.00 (E05)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-20.2	-14.7	-25.5	1.3	-19.9	-19.9	1.8	-14.3	27.0	21.9	24.3	19.9	4.5	0	0.605

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.8	96.6	67.5	93.0	66.5	89.6	65.6	70.5	87.6	69.1	86.2	67.7	84.3	8.8	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.7	95.2	74.8	64.0	89.5	73.9	62.2	84.1	73.3	34.4	87.5	33.2	86.3	32.1	83.9	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.6	20.2	18.3	-26.2	101.3	6.8	3.0	-31.1	103.4	-35.1	105.1	-38.9	106.8	-43.8	109.0		
(5)			-26.5	74.3	6.7	2.4	-31.3	76.0	-35.2	77.4	-39.0	78.7	-43.8	80.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	43.3	11.4	11.8	25.6	46.8	60.3	68.9	72.9	70.7	58.4	44.3	29.7	17.0		
	DBStd	24.56	13.37	13.20	11.02	10.09	8.35	7.94	6.46	7.82	8.47	8.56	10.92	12.87		
	HDD50	5092	1196	1070	759	176	13	1	0	0	20	222	611	1024		
	HDD65	8628	1661	1490	1223	549	192	48	10	33	231	642	1060	1489		
	CDD50	2655	0	0	1	79	334	569	711	643	272	45	1	0		
	CDD65	715	0	0	0	2	48	166	255	211	33	0	0	0		
	CDH74	9728	0	0	0	64	790	2247	3251	2820	548	8	0	0		
	CDH80	4427	0	0	0	8	269	1048	1539	1376	187	0	0	0		
	WSAvg	8.2	9.0	9.3	9.4	9.3	8.1	7.5	7.1	7.2	7.2	7.9	8.4	8.6		
	PrecAvg	14.4	1.1	0.9	1.1	1.0	1.3	1.4	1.5	1.1	1.1	1.5	1.2	1.1		
	PrecMax	19.4	2.5	1.5	2.5	1.9	3.3	3.9	3.8	3.2	2.8	3.4	3.2	2.8		
(15) Precipitation	PrecMin	8.1	0.3	0.3	0.0	0.1	0.0	0.1	0.2	0.2	0.1	0.1	0.1	0.1		
	PrecStd	3.0	0.5	0.3	0.7	0.5	0.7	0.8	1.1	0.8	0.7	0.8	0.8	0.6		
	DB	35.8	35.9	59.3	80.6	90.2	99.0	100.3	100.6	90.8	74.6	55.2	41.4			
	MCWB	34.8	34.5	47.8	58.8	64.5	67.8	68.4	67.9	61.6	55.1	50.7	39.8			
	2% DB	34.0	34.1	48.2	75.1	86.4	95.0	95.3	96.4	85.6	69.5	49.8	36.0			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	33.2	33.2	42.3	56.4	63.0	67.3	67.6	67.1	60.5	53.0	46.2	34.9			
	5% DB	32.4	33.1	41.8	69.9	82.8	90.6	91.8	92.2	80.6	63.8	46.1	34.2			
	MCWB	32.0	32.3	38.3	54.0	62.1	66.1	67.4	65.9	59.0	50.8	43.9	33.4			
	10% DB	30.5	31.2	37.7	64.6	78.4	86.2	88.6	87.9	75.4	58.4	42.6	33.1			
	MCWB	29.9	30.5	35.2	51.0	60.3	65.2	66.4	65.2	57.6	49.4	40.7	32.5			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	35.1	34.9	48.1	61.2	68.0	72.5	72.6	71.3	65.7	59.5	52.0	40.3			
	MCDB	35.3	35.3	58.6	77.2	84.3	88.8	88.4	90.1	83.3	68.6	54.5	41.3			
	2% WB	33.4	33.4	42.6	57.8	65.4	69.8	70.8	69.3	62.9	55.1	46.7	35.3			
	MCDB	33.7	33.8	47.3	71.4	81.6	86.8	87.1	88.6	77.4	64.3	48.6	35.8			
	5% WB	32.1	32.2	38.7	55.0	63.4	68.1	69.3	68.0	61.0	52.6	43.8	33.6			
(31) Mean Daily Temperature Range	MCDB	32.5	32.9	41.6	67.1	79.5	84.9	85.9	86.7	75.7	61.0	45.5	34.0			
	10% WB	29.9	30.2	35.5	52.2	61.2	66.5	68.0	66.3	59.0	50.1	40.5	32.2			
	MCDB	30.6	31.1	37.2	63.2	75.5	82.4	83.8	83.9	71.8	56.7	42.1	32.8			
	5% DB	14.1	16.7	15.8	21.7	24.2	25.2	24.8	26.3	24.6	19.1	12.5	13.0			
	MCDBR	9.3	11.2	17.4	28.7	31.2	31.1	29.5	31.4	32.2	28.2	15.2	9.6			
(36) Clear-Sky Solar Irradiance	MCWBR	9.4	11.0	12.3	14.2	13.6	12.1	10.5	11.8	14.4	15.8	12.4	9.0			
	5% WB	9.5	11.6	16.8	26.5	28.4	26.7	25.8	28.2	27.7	23.7	13.5	8.6			
	MCDBR	9.7	11.6	12.2	14.4	13.9	11.7	10.0	11.7	14.2	15.1	11.7	8.5			
	taub	0.314	0.337	0.386	0.439	0.417	0.408	0.414	0.429	0.383	0.358	0.337	0.312			
	taud	2.057	2.013	2.010	2.070	2.222	2.308	2.297	2.227	2.317	2.312	2.239	2.108			
(42) All-Sky Solar Radiation	Ebn at Noon	208	239	250	252	266	269	265	253	253	235	204	189			
	Edh at Noon	26	36	44	47	42	39	39	40	33	27	22	22			
	RadAvg	340	709	1100	1446	1884	2058	2003	1708	1190	655	294	216			
(45)	RadStd	82	76	166	136	133	172	139	117	141	90	40	55			

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	DBAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)	DBAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page